Strategic Data Source Acquisition and Integration for the "Fessi" Waste Management Knowledge Graph

Executive Summary: The Data Architecture of Sustainable Disposal

The development of "Fessi," a Graph-RAG (Retrieval-Augmented Generation) system for a German university, represents a sophisticated challenge in semantic modeling and data integration. The core objective—automating waste disposal guidance for queries ranging from "old paint cans" to "lab chemicals"—requires a knowledge graph that bridges the gap between colloquial user intent and rigid regulatory frameworks. Unlike general information retrieval, the domain of waste management operates at the intersection of material science, chemical safety, and logistical engineering. A successful resolution to a student's query about a "broken beaker" involves determining the material composition (borosilicate glass), its contamination state (hazardous chemical residue), and the physical constraints of the disposal infrastructure (sharps containers versus glass recycling bins).

This report details an exhaustive investigation into the data sources required to populate the "Fessi" knowledge graph, specifically focusing on the critical layer of **Material Properties**. Our research indicates that no single "super-database" exists to cover the requisite spectrum of density, toxicity, flammability, and biodegradability. Instead, the waste management domain is characterized by a "data dichotomy": scientific databases offer high-precision molecular data (e.g., crystal density), while logistical datasets provide low-precision bulk data (e.g., dumpster weight). The "Fessi" system must therefore function as a data fusion engine, synthesizing authoritative chemical safety data from sources like **PubChem** and **ECHA** with pragmatic engineering data from the **EPA** and **MatWeb**, all while adhering to the German **AVV (Abfallverzeichnis-Verordnung)** regulatory context.

The findings presented herein prioritize the extraction of **density data** as a critical vector for weight estimation, which directly impacts disposal cost modeling. Furthermore, we identify the **European Chemicals Agency (ECHA)** database as the indispensable "source of truth" for hazard classification within the EU, superseding broader international datasets for compliance purposes. The report outlines a tiered integration strategy, recommending a "federated" knowledge graph architecture where static physical properties are augmented by dynamic regulatory lookups, ensuring that "Fessi" remains both physically accurate and legally compliant.

---

1. The Physics and Chemistry of Waste: A Data Requirements Analysis

Before examining specific repositories, it is essential to deconstruct the physical and chemical attributes that drive disposal logic within the Knowledge Graph. The "Fessi" ontology relies on three layers—Core Entities, Logic, and Infrastructure—but the decision-making engine resides entirely in the attributes of the WasteItem. The semantic resolution of a disposal query is fundamentally a property-matching exercise against a set of exclusion rules.

1.1 The Density Paradox in Waste Logistics

The most urgent requirement identified is density data for weight estimation. However, "density" is a polymorphic term in this domain, leading to potential catastrophic errors in weight calculation if not properly distinguished.

- **True Density (ρ_{true}):** This is the density of the solid material itself, excluding all pores and voids. For example, the true density of soda-lime glass is approximately $2,500 \text{ kg/m}^3$. This value is critical for the "Safety Identification" logic layer; knowing the true density helps separate materials in float/sink tanks during recycling processes. Scientific databases like the **Materials Project** and **NIST** excel at providing this granular data [1, 2].
- **Bulk Density (ρ_{bulk}):** This is the mass of the material divided by the total volume it occupies, including air gaps between particles or items. A bin full of empty glass bottles has a bulk density significantly lower than solid glass—often ranging from 300 to 600 kg/m^3 depending on whether the glass is whole or crushed [3]. For the "Infrastructure" layer of the graph (Container, Building), ρ_{bulk} is the only relevant metric. Using ρ_{true} to estimate the weight of a dumpster would result in a cost overestimation of 400-500%.
- **Compaction Factors:** Waste is rarely static. Cardboard behaves differently when "loose" versus "baled." The data sources selected must therefore provide not just a single scalar value for density, but a range of values contingent on the material's physical state (loose, compacted, shredded).

1.2 Chemical Safety and the "Hazard" Attribute

For laboratory waste, the graph must support complex inferential reasoning. A query regarding "Acetone" must trigger a DisposalRule that checks for flammability. This requires data on **Flash Point** and **Autoignition Temperature**. Furthermore, the German regulatory environment relies on the GHS (Globally Harmonized System) standards implemented via the CLP Regulation (Classification, Labelling and Packaging).

- **GHS Signal Words:** "Danger" vs. "Warning."
- **H-Phrases (Hazard Statements):** Codes like H200 (Unstable Explosive) or H300 (Fatal if swallowed) are the primary keys for mapping items to hazardous waste streams

(Sonderabfall) [4].
- **Reactivity:** The graph needs isIncompatibleWith relationships. Oxidizers must be flagged to prevent storage with flammables. This requires deep chemical property data found in **PubChem** and **ECHA** dossiers [5, 6].

1.3 Biodegradability and the "Time" Dimension

The transition to sustainable campuses places a heavy emphasis on organic waste diversion. However, "biodegradable" is often a marketing term rather than a technical specification. The graph requires precise data on:

- **Half-life:** The time required for 50% degradation in specific environments (soil, marine, anaerobic digester) [7].
- **Certification Standards:** Logic rules must differentiate between Home Compostable (lower temperature degradation) and Industrial Compostable (e.g., DIN EN 13432, requiring 55°C+).

---

2. Primary Chemical and Safety Data Sources

The foundation of the "Fessi" knowledge graph, particularly for the university's laboratory and hazardous waste sectors, must be built upon authoritative chemical repositories. These sources provide the "molecular truth" of a substance.

2.1 PubChem (National Institutes of Health)

Source Identification

- **Name:** PubChem
- **Provider:** National Center for Biotechnology Information (NCBI), National Library of Medicine (NLM), NIH.
- **URL:** https://pubchem.ncbi.nlm.nih.gov/
- **Type:** Open Chemistry Database [8].

Data Format & Access

- **API:** PubChem offers the **PUG REST (Power User Gateway)**, a sophisticated API designed for programmatic access. It supports highly structured queries without requiring an API key, although it enforces dynamic rate limiting (typically capped at 5 requests per second) to ensure service stability [9].
- **Endpoints:** The API is RESTful and hierarchical. Access patterns typically involve resolving a name to a Compound ID (CID) and then requesting specific data "views."
  - Input: GET /rest/pug/compound/name/{ChemicalName}/cids/JSON
  - Property Retrieval: GET /rest/pug/compound/cid/{CID}/property/{PropertyTags}/JSON
- **Data Formats:** JSON, XML, CSV, and SDF (Structure-Data File).

Coverage & Quality

- **Record Count:** The database contains over 111 million unique chemical structures and nearly 300 million substance records [10].
- **Geo Scope:** Global. While hosted in the US, it aggregates data from international contributors, including German/European sources (e.g., Thomson Reuters, Thieme Chemistry).
- **Quality:** The "Laboratory Chemical Safety Summary" (LCSS) view is curated and high-fidelity. However, general property data is aggregated from multiple depositors, which can sometimes lead to conflicting values (e.g., slightly different boiling points from different vendors).

Licensing

- **Commercial Use:** PubChem data is largely in the public domain as a US government work. There are no restrictive licenses for the core data, making it safe for integration into the university's project without royalty fees [8].

Relevance Assessment (Score: 5/5 - Critical)
PubChem is the "spine" of the chemical knowledge layer. Its synonym handling is unmatched, allowing "Fessi" to understand that "Bleach," "Sodium Hypochlorite," and "NaOCl" are the same entity [11].
Integration Notes & Data Mapping
For "Fessi," the generic compound summary is too noisy. The integration strategy should strictly utilize the LCSS (Laboratory Chemical Safety Summary) API view. This specific view filters the dataset to present only the health and safety-critical information required for waste handling.

Fessi Attribute	**PubChem API Field / Path**	**Logic Application**
HazardClassification	GHS Classification (Section: Safety and Hazards)	Maps directly to "Red Flag" warnings in the chatbot.
FlashPoint	Physical Properties -> Flash Point	Used to categorize flammable liquids (AVV 07 01 04*).
Toxicity_LD50	Toxicity -> Acute Toxic Effects	Quantifies the "danger" level for transport rules.
MolecularFormula	Molecular Formula	Disambiguation of isomers.

Synonyms	Synonyms	NLP Training Data: Populates the matches list for Entity Recognition.

Similar/Alternative Sources

- **ChemSpider:** Excellent alternative, but API access is more restrictive and often requires a paid license for high-volume commercial use.
- **ChEMBL:** Better for bioactivity, less focused on physical waste properties.

2.2 ECHA (European Chemicals Agency) - REACH Database

Source Identification

- **Name:** ECHA CHEM / REACH Database
- **Provider:** European Chemicals Agency (EU).
- **URL:** https://chem.echa.europa.eu/ [6].
- **Type:** Regulatory Database.

Data Format & Access

- **API:** ECHA does *not* offer a high-performance, real-time REST API comparable to PubChem.
- **Bulk Access:** Data is available via **IUCLID** (International Uniform ChemicaL Information Database) exports. ECHA provides specific downloadable datasets, such as the *C&L Inventory* (Classification and Labelling), which is updated regularly [12].
- **Format:** IUCLID XML, CSV extracts.

Coverage & Quality

- **Geo Scope: European Union**. This is the single most important factor for a German university.
- **Quality:** Regulatory Grade. The data here represents the legal consensus on a chemical's hazard status within the EU [13].
- **Record Count:** Covers all substances registered under REACH (>25,000 unique substances) and the C&L Inventory (>150,000 entries).

Licensing

- **Attribution:** ECHA data is free to use but requires strict source attribution ("Source: European Chemicals Agency, http://echa.europa.eu/") [14].
- **Commercial Use:** Permitted, provided the legal notice is respected.

Relevance Assessment (Score: 5/5 - Compliance Essential)
While PubChem provides the physics of a chemical, ECHA provides the law. If ECHA classifies

a solvent as a "Carcinogen 1B," the university is legally obligated to treat it as hazardous waste, regardless of what other databases might suggest. This database governs the mapping to AVV Codes (e.g., identifying "Substances of Very High Concern" - SVHC) [15].
Integration Notes & Data Mapping
Due to the lack of a real-time API, the recommended integration pattern is Batch Ingestion. The "Fessi" team should download the C&L Inventory and Candidate List of SVHCs to build a local lookup table.

Fessi Attribute	ECHA Data Point	Logic Application
REACH_Registration	Registration Dossier	Verifies if the substance is legally registered for use/disposal in the EU.
SVHC_Status	Candidate List	**Critical Logic:** IF True, strict segregation is mandatory.
CLP_Labeling	C&L Inventory	Provides the official H-phrases (e.g., H411 - Toxic to aquatic life) for labelling.
EC_Number	EC Number	The primary identifier for European regulatory forms.

Gap Analysis
ECHA data is document-heavy and text-heavy. Extracting discrete physical values (like exact density) from the IUCLID dossiers is technically complex compared to PubChem's structured JSON. Use ECHA for classification and PubChem for properties.

2.3 EPA EPI Suite™ (Estimation Programs Interface)

Source Identification

- **Name:** EPI Suite™
- **Provider:** U.S. Environmental Protection Agency (EPA) & SRC, Inc.
- **URL:** https://www.epa.gov/tsca-screening-tools/epi-suitetm-estimation-program-interface [16].
- **Type:** QSAR Modeling Software.

Data Format & Access

- **Access:** This is downloadable software (Windows Executable), not a web API.
- **Mechanism:** It uses SMILES strings to *estimate* properties using QSAR (Quantitative Structure-Activity Relationship) models [17].
- **Web Beta:** A beta web version exists (https://episuite.dev/EpiWebSuite/), but for a robust Knowledge Graph, the offline calculation engine is preferred [18].

Coverage & Quality

- **Focus:** Environmental Fate and Transport.
- **Unique Value:** It fills the "Data Void." For thousands of chemicals, experimental data for biodegradability does not exist. EPI Suite provides scientifically valid estimates.

Licensing

- **Commercial Use:** Public Domain (US Gov). The software is free to download and use [16].

Relevance Assessment (Score: 3/5 - Gap Filling)
Essential for the Biodegradability and Environmental Persistence requirements of the user request. Standard databases often lack "half-life in soil" data.
Integration Notes & Data Mapping
This tool should be used "offline" to pre-calculate properties for the most common 500-1,000 lab chemicals.

Fessi Attribute	EPI Suite Module	Logic Application
Biodegradability_Predicted	BIOWIN	Predicts if a substance degrades rapidly or persists (Organic vs. Chemical waste).
Bioaccumulation_Factor	BCFBAF	Determines risk to aquatic environments.
Kow	KOWWIN	Octanol-water partition coefficient; predicts movement in groundwater.

---

3. Physical & Engineering Data Sources

Moving beyond pure chemicals, the "Fessi" graph must handle solid waste streams—metals,

plastics, glass, and construction debris.

3.1 The Materials Project

Source Identification

- **Name:** The Materials Project
- **Provider:** Lawrence Berkeley National Laboratory (LBNL).
- **URL:** https://next-gen.materialsproject.org/ [19].
- **Type:** Computational Materials Science Database.

Data Format & Access

- **API:** Excellent, modern API support via the mp-api Python client.
- **Auth:** Requires an API Key (free for academic/research use upon registration) [20].
- **Endpoints:**
  - Summary: mpr.materials.summary.search()
  - Thermo: mpr.materials.thermo.search()
- **Format:** JSON structure containing computed properties [21].

Coverage & Quality

- **Record Count:** >150,000 inorganic compounds.
- **Focus: Inorganic Materials** (Crystals, Salts, Metals, Ceramics, Battery compounds). It is *weak* on organic polymers (plastics).
- **Quality:** Data is computed via Density Functional Theory (DFT). It represents "theoretical perfect crystals" rather than "dirty waste" [22].

Licensing

- **Attribution:** Creative Commons Attribution 4.0 International (CC BY 4.0). Requires citation of the Materials Project in any public output [23].

Relevance Assessment (Score: 4/5 - Specific Utility)
This source is highly specific. It is not useful for "paper waste" but is the definitive source for E-Waste (Semiconductors), Battery Materials, and Lab Ceramics.
Integration Notes & Data Mapping

Fessi Attribute	Materials Project Field	Logic Application
TrueDensity	density	Calculated density of the crystal structure [2].
Composition	elements	Identifies heavy metals (e.g., Cd, Pb) for toxicity

		rules.
Conductivity_BandGap	band_gap	Distinguishes between metals (conductor) and ceramics (insulator).
Stability	formation_energy_per_atom	Predicts chemical stability.

Gap Identified: The density values here are ρ_{true}. They must NOT be used for bin volume calculations, only for material identification.

3.2 MatWeb

Source Identification

- **Name:** MatWeb
- **Provider:** MatWeb, LLC.
- **URL:** https://www.matweb.com/ [24].
- **Type:** Engineering Materials Database.

Data Format & Access

- **API:** MatWeb does **not** offer a public, free API. It is a proprietary database.
- **Access:** Web interface for search. Premium membership allows for data export to FEA/CAD formats (e.g., Excel, XML) [25].
- **Terms:** Scraping is explicitly prohibited.

Coverage & Quality

- **Record Count:** >185,000 data sheets [26].
- **Focus: Polymers** (Thermoplastics, Thermosets), Metals, Ceramics.
- **Quality:** Aggregated from manufacturer data sheets. This is "real world" engineering data, unlike the theoretical data from Materials Project.

Licensing

- **Commercial Use:** Proprietary. Usage typically requires a paid license for integration or bulk export. "The MatWeb™ materials database content is licensed, not sold" [27].

Relevance Assessment (Score: 3/5 - High Value / High Friction)
MatWeb fills the massive gap in Polymer Data left by PubChem and Materials Project. It is the best source for distinguishing the properties of HDPE vs LDPE vs PET.
Integration Recommendation

Do not attempt to build a real-time connector. Instead, identifying the "Top 50" most common waste polymers (Nylon, ABS, Polystyrene, etc.) and manually curating their density/thermal properties from MatWeb into the "Fessi" graph is the most compliant and stable approach.

Fessi Attribute	MatWeb Field	Logic Application
Material_Density	Density	Engineering density ranges (e.g., 0.9 - 0.95 g/cc for PE).
MeltingPoint	Melting Point	Determines suitability for autoclaving lab waste.
TensileStrength	Tensile Strength	Proxy for "shreddability" in waste processing.

---

4. Waste Logistics Sources (Density & Volume)

This section addresses the user's "high priority" requirement for density data. The scientific sources above provide ρ_{true}. This section provides ρ_{bulk}—the data required to answer "How heavy is this dumpster?"

4.1 EPA Standard Volume-to-Weight Conversion Factors

Source Identification

- **Name:** Volume-to-Weight Conversion Factors for Solid Waste
- **Provider:** U.S. EPA (Office of Resource Conservation and Recovery).
- **Format:** PDF Report (April 2016 Update) [28].
- **URL:** https://www.epa.gov/smm/volume-weight-conversion-factors-solid-waste

Data Format & Access

- **Access:** Downloadable PDF. No API.
- **Structure:** Tables listing waste types (e.g., "Glass - Whole Bottles") against conversion factors (lbs/cu yd).

Coverage & Quality

- **Granularity:** Excellent. It distinguishes between distinct states of matter:
  - *Loose* (thrown in a bin).
  - *Compacted* (in a truck).
  - *Baled* (ready for recycling).
- **Relevance:** It covers standard municipal waste streams: Paper, Glass, Plastic, Metals,

Organics, Construction Debris.

Licensing

- **Status: U.S. Public Domain**. Free to use and reproduce.

Relevance Assessment (Score: 5/5 - Logistics Essential)
This is the "Rosetta Stone" for the infrastructure layer of the Knowledge Graph. It allows "Fessi" to convert the user's "3 bags of trash" into "15 kg of waste."
Integration Notes & Data Mapping
The data must be parsed from the PDF into a structured Lookup Table (CSV/JSON) stored within the graph's WasteStream nodes.

Fessi Attribute	EPA Data Point	Example Value
BulkDensity_Loose	Volume-to-weight factor (Loose)	200 lbs/yd³ (Cardboard)
BulkDensity_Compacted	Volume-to-weight factor (Compacted)	500 lbs/yd³ (Cardboard)
Material_State	Material Form	"Whole", "Flattened", "Crushed"

Gap Analysis: The data uses US Imperial Units (lbs/yd³). A strict conversion layer to Metric (kg/m^3) is required for the German context.

4.2 UK WRAP (Waste and Resources Action Programme)

Source Identification

- **Name:** Material Bulk Densities
- **Provider:** WRAP (UK Charity).
- **URL:** https://www.wrap.ngo/resources/report/material-bulk-densities [29].

Relevance
Acts as the metric counterpart to the EPA dataset. It provides bulk density data specifically for UK/European waste collection vehicles and containers. This aligns closer to the German logistics model than US data.

---

5. German-Specific Data Sources: The Context Layer

For a German university, global data must be filtered through local regulations.

5.1 AVV (Abfallverzeichnis-Verordnung)

Source Identification

- **Name:** Abfallverzeichnis-Verordnung (AVV) / European Waste Catalogue (EWC).
- **Provider:** German Federal Ministry of Justice / EU Commission.
- **URL:** https://www.gesetze-im-internet.de/avv/ [30].

Data Format & Access

- **Format:** XML, PDF, HTML text.
- **Structure:** A 6-digit hierarchical code system (e.g., 20 01 21 = fluorescent tubes).

Relevance Assessment (Score: 5/5 - Taxonomy Core)
The AVV is not a property database; it is the Ontology. Every WasteItem in "Fessi" must map to an AVV code. This code determines the legal disposal route.

- **Hazard Flag:** Any code with an asterisk (*) is legally hazardous (e.g., 20 01 35* discarded electrical equipment containing hazardous components).

Integration Strategy
Ingest the full AVV XML list. Use this as the "skeleton" of the Knowledge Graph. Attach physical properties (from PubChem/EPA) to these AVV nodes.

5.2 Abfallkalender & Municipal APIs

Source Identification

- **Providers:** Abfall+, MyMuell, local city portals (e.g., Würzburg, Berlin Open Data) [31, 32].
- **Type:** Logistics/Service APIs.

Relevance (Score: 4/5 - User Context)
While not providing "material properties," these sources provide "Location Data" (requested in the prompt). They answer: "When is the pickup?"

- **Integration:** Many German districts use **Abfall+** or **MyMuell** backends which often expose JSON or ICS endpoints [33]. "Fessi" can query these to provide real-time collection schedules based on the user's location.

---

6. Emerging Domains: Batteries & Bioplastics

6.1 Battery Failure Databank (NREL/NASA)

Source Identification

- **Name:** Battery Failure Databank
- **Provider:** NREL & NASA.

- **URL:** https://www.nrel.gov/transportation/battery-failure.html [34].
- **Format:** Excel/CSV Downloads.

Relevance (Score: 4/5 - Safety Critical)
Lithium-ion batteries are a major fire risk in waste streams.
- **Data:** Provides **Heat Output**, **Mass Ejection**, and **Thermal Runaway** profiles.
- **Application:** If a user asks, "Can I throw this swollen laptop battery in the bin?", the graph can access the high-risk profile of Li-ion cells to trigger a "STOP - EXPLOSION RISK" warning.

6.2 Bioplastics Certification (DIN CERTCO)

Source Identification

- **Name:** DIN CERTCO Database
- **Provider:** TÜV Rheinland / DIN.
- **Standard:** DIN EN 13432 (Compostability) [35].

Relevance (Score: 5/5 - Logic Critical)
Distinguishing "Bio-based" (made from plants but not degradable) from "Biodegradable" (compostable) is notoriously difficult.
- **Data:** Registers of certified products.
- **Gap:** No API. Requires scraping or manual lookup of certification numbers (e.g., "7P0023").

---

7. Integrated Architecture Recommendation

To fulfill the "Fessi" project goals, we recommend a **Federated Knowledge Graph Architecture** that fuses these disparate sources.

7.1 The Data Ingestion Pipeline

1. **The Regulatory Skeleton (Layer 1):**
   - Import the **AVV (Abfallverzeichnis)** XML. Create nodes for all 800+ waste codes.
   - *Source:* German Legal Text Archives.
2. **The Chemical Flesh (Layer 2):**
   - For every chemical entity, query **PubChem** (PUG REST) for CIDs.
   - Ingest the **ECHA C&L Inventory** to flag hazards (H-phrases).
   - *Result:* An entity Acetone that knows it is "Flammable" (PubChem) and "Eye Irritant Category 2" (ECHA).
3. **The Physical Body (Layer 3):**
   - Map waste categories (Paper, Glass) to **EPA SMM** bulk density tables.
   - *Result:* The graph knows that Cardboard weighs 120 kg/m^3 when loose.
4. **The Engineering Patch (Layer 4):**

    - Use **MatWeb** data for specific polymer properties.
    - Use **Materials Project** for identifying inorganic lab waste components.

7.2 Addressing Gaps

- **Composite Objects:** No database lists the material breakdown of a specific "Nike Air Max" or "iPhone 13." "Fessi" must use a **Decomposition Rule** (e.g., Sneaker -> ``) and aggregate the properties of the components.
- **Electrical Conductivity:** While **Materials Project** provides band-gap data for pure crystals, identifying the conductivity of mixed e-waste remains a heuristic estimation based on metal content.
- **German Synonyms:** Linking *"Styropor"* to *"Expanded Polystyrene"* requires a robust SKOS (Simple Knowledge Organization System) synonym layer, as most scientific APIs are Anglocentric.

By implementing this multi-source strategy, "Fessi" can evolve from a simple FAQ bot into a sophisticated environmental compliance engine, capable of reasoning through the physics, chemistry, and laws of waste management.

---

8. Detailed Source Profile Matrix

Source Name	Primary Domain	Data Access	Licensing	Key "Fessi" Attribute
PubChem	Chemical Safety	API (REST), JSON	Public Domain	FlashPoint, Toxicity
ECHA CHEM	EU Regulation	Bulk Download (IUCLID)	Attribution Req.	SVHC_Status, CLP_Labeling
EPA SMM	Waste Logistics	PDF / Static Tables	Public Domain	BulkDensity (Volume-to-Weight)
Materials Project	Inorganic Science	API (Python Client)	CC BY 4.0	TrueDensity, Composition
MatWeb	Polymers/Engineering	Web / Proprietary	Commercial	TensileStrength, PolymerDensit

				y
NREL Battery DB	Li-ion Safety	CSV Download	Public Domain	ThermalRunawayProfile
NIST WebBook	Thermochemistry	Web / Scraping	Public Domain	EnthalpyOfCombustion
Abfall+ / MyMuell	Local Logistics	JSON / ICS API	Var. (Open Data)	CollectionDate, BinType
EPI Suite	Biodegradability	Software (QSAR)	Public Domain	BiodegradationHalfLife

This matrix serves as the procurement guide for the "Fessi" data engineering team.